

OTP2.0 v4 — Investment Thesis

Active-management overlay on a tech-tilted equity sleeve

STRATEGY OVERVIEW

OTP2.0 v4 combines two pillars:

1. Security Selection: a 54-stock universe spanning old-economy mega-caps (MSFT, JNJ, PG, JPM, ...) and modern growth/tech leaders (NVDA, META, AMZN, AVGO, TSLA, AMD, ...). Every 3 years, stocks are ranked on a 5-factor composite score:
 - F1 12-month momentum 30%
 - F2 % days above 50-day SMA 20%
 - F3 3-yr CAGR / volatility 20%
 - F4 Drawdown resilience ($1/|\text{MaxDD}|$, 5yr) 10%
 - F5 252-day relative strength vs SPX 20%The top 7 stocks form an equal-weighted "cohort" held for 3 years.
2. Timing Overlay (OTPl.0, locked engine): a daily two-bucket (invested/cash) allocator that scales exposure using realized volatility, trend (SMA50/100/200), VIX-based risk triggers, and a gradual reload mechanism after risk-off periods.

REGIME-AWARE CONFIGURATION (v4's key addition)

Each cohort's timing configuration is set ex-ante (no lookahead) based on two signals measured at the selection date:

- Composite-tertile: cohorts whose mean factor-composite score is in the top third get a "Momentum" config (higher vol target, faster reloads); bottom third get a tighter "Defensive" config; middle third use the default OTPl.0 config.
- CAPE valuation override: if the Shiller CAPE ratio at the selection date is ≥ 30 (roughly the top decile of its 1881-present history), the cohort is forced to the Defensive config REGARDLESS of its momentum tertile. This flags 1999, 2020, and 2026 as "frothy" markets where momentum-based optimism historically preceded sharp corrections.

RESULT vs. THE BAR

Target bar: chained Sharpe > 1.0 OR CAGR $> 15\%$, with a "significant margin" over the market (SPX B&H: Sharpe 0.322, CAGR 9.17%).

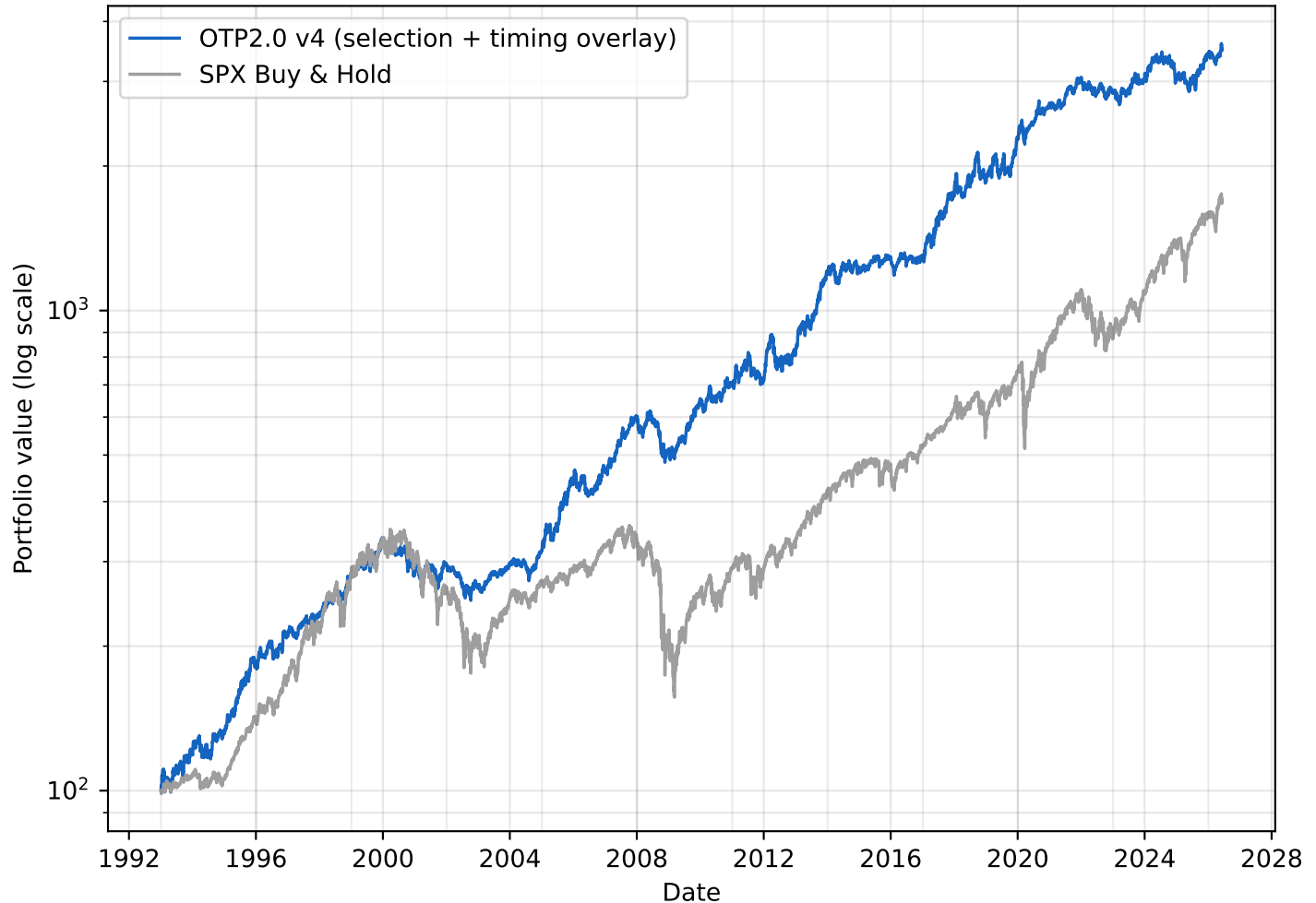
v4 chained (1993-2026, hindsight-free, sequential cohorts):

Sharpe	0.654	($\approx 2.0\times$ SPX)
CAGR	11.25%	(SPX 9.17%, +2.1pp)
MaxDD	-26.1%	(SPX -56.8%, less than half)

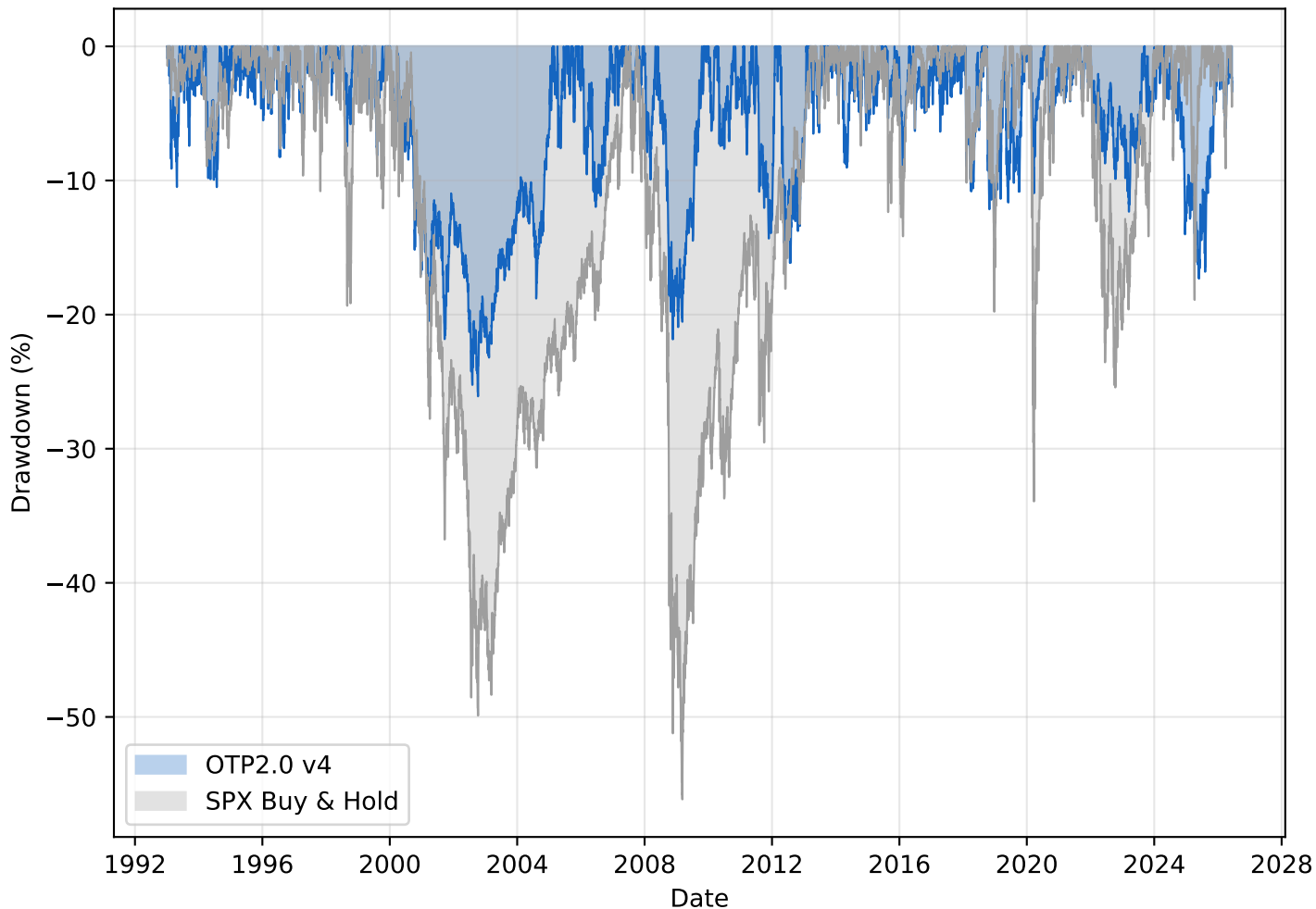
- > Bar NOT reached. v4 delivers a real, repeatable risk-adjusted edge (roughly double SPX's Sharpe) and a materially shallower drawdown, but the absolute return edge over SPX (+2.1pp CAGR) is modest, not the "significant margin" the bar requires.

This thesis presents v4 as the best-performing of four iterations tested (v1-v4), with full transparency on its regime behavior and where its edge comes from.

Chained Equity Curve (Hindsight-Free, 1993-2026) Starting value = 100, log scale

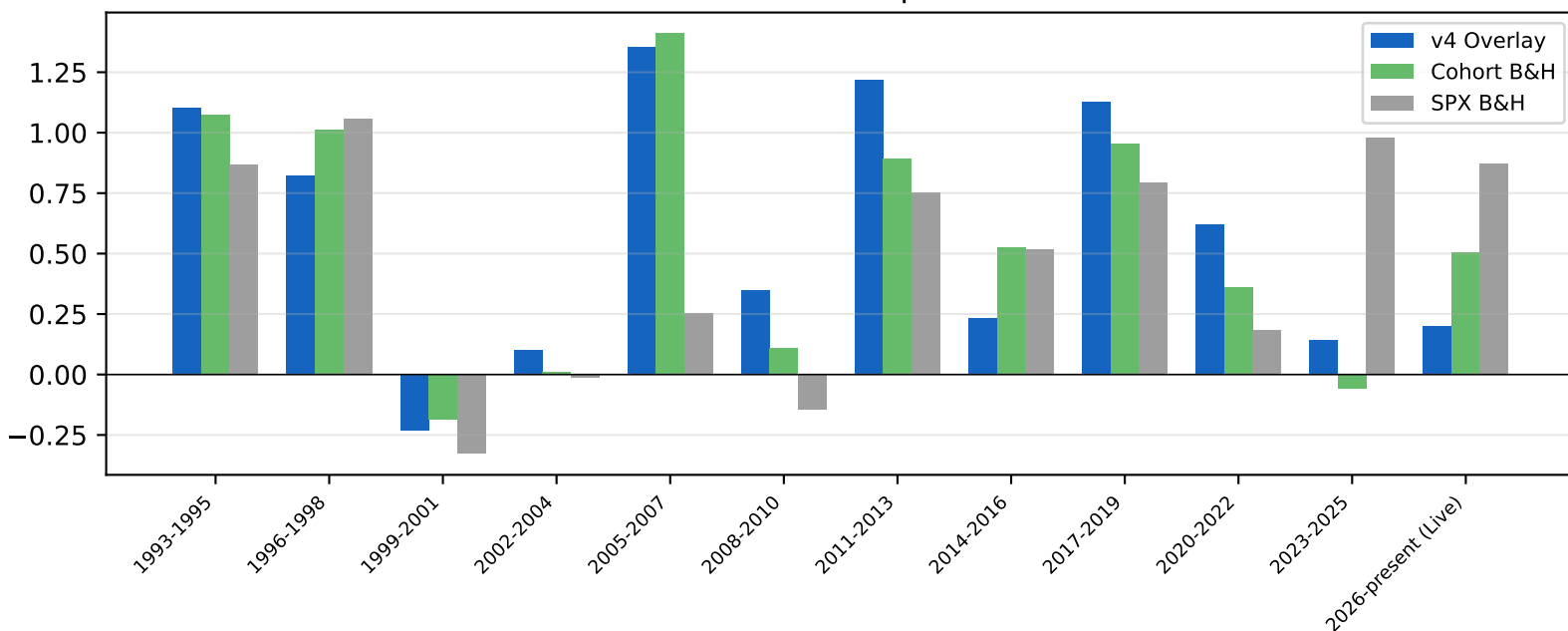


Underwater Equity (Drawdown from Running Peak) v4 MaxDD -26.1% vs SPX MaxDD -56.8%

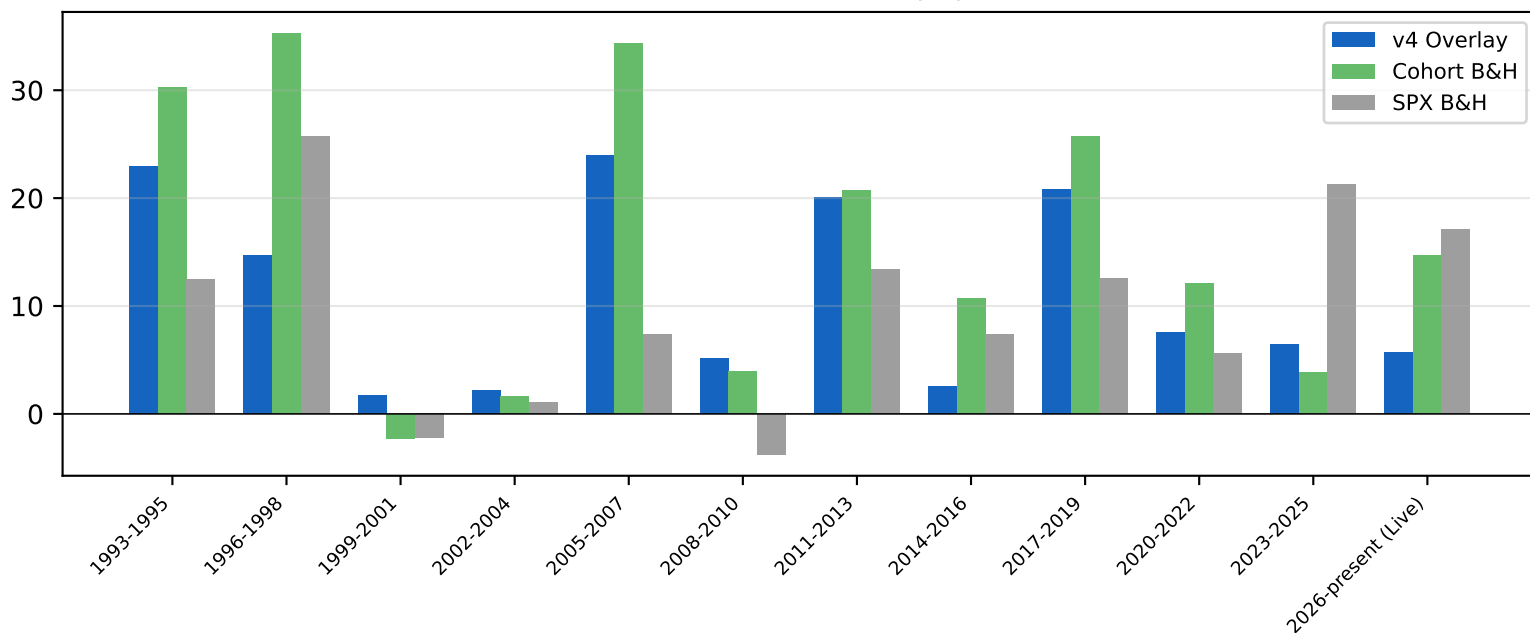


Per-Cohort Performance: v4 Overlay vs. Cohort Buy&Hold vs. SPX

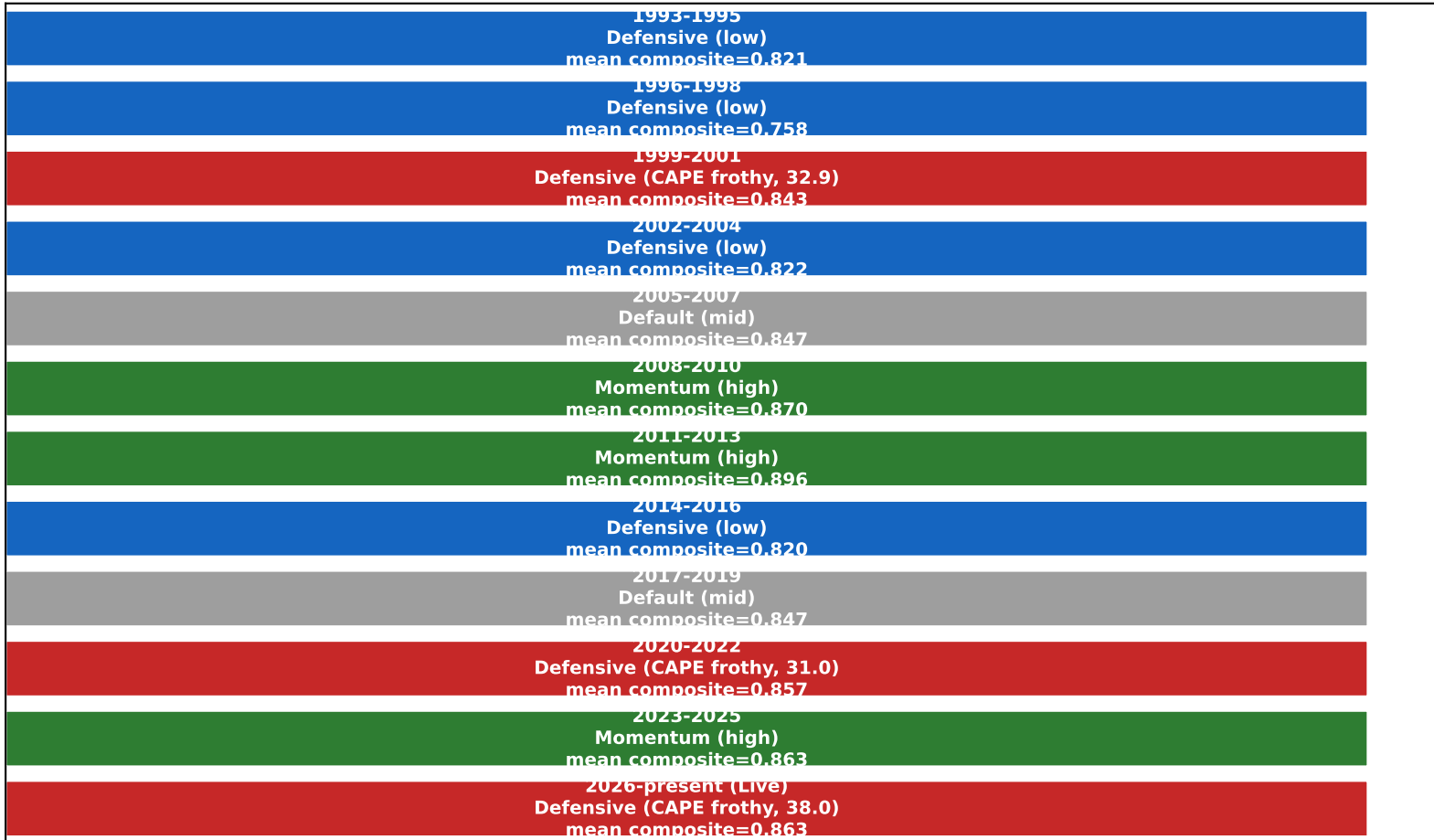
Per-Cohort Sharpe Ratio



Per-Cohort CAGR (%)



Regime-Tilt Timeline: Timing Configuration Assigned per Cohort
(green=Momentum, gray=Default, blue=Defensive-tertile, red=Defensive-CAPE-frothy)



Target bar: Sharpe > 1.0 OR CAGR > 15% -> NOT REACHED by any variant
v4 is the best risk-adjusted result (~2x SPX Sharpe, less than half SPX's drawdown)

	Variant	Sharpe	CAGR	MaxDD
v1	(original, 5yr cohorts, to	0.625	8.89%	-17.7%
v2 (50-	stock universe, fixed thr	0.642	13.12%	-52.0%
v3	(regime tertiles, tech t	0.617	11.06%	-33.4%
v4	(+ CAPE froth override	0.654	11.25%	-26.1%
	SPX Buy & Hold	0.322	9.17%	-56.8%